

Problem 1.*Solution:*

a) Using the Monte Carlo estimation, we have:

$$\int_0^1 e^{e^x} = E[e^{e^U}] \quad (1)$$

$$g(U) = e^{e^U} \quad (2)$$

$$X = \frac{1}{n}[g(U_1) + \dots + g(U_n)] \quad (3)$$

pseudo code:

```

1  repeat i = 1..n:
2      U = Uniform(0,1)
3      X = 1/n * [sum(g(U_1), ..., g(U_n))]
4  output X

```

Converting this pseudo-code to python code and running for 1000 trials gives the estimate of the integral to be: 6.3972

b) Using the Monte Carlo estimation, we have:

$$\int_{-2}^2 e^{x+x^2} = (2 - (-2))E(e^{x+x^2}) \quad (4)$$

$$= 4E(e^{x+x^2}) \quad (5)$$

$$g(U) = e^{x+x^2} \quad (6)$$

$$X = 1/n * [g(U_1) + \dots + g(U_n)] \quad (7)$$

pseudo code:

```

1  repeat i = 1..n:
2      U = Uniform(0,1)
3      V = -2 + 4U
4      X = 1/n * [sum(g(V_1), ..., g(V_n))]
5  output X

```

Converting this pseudo-code to python code and running for 1000 trials gives the estimate of the integral to be: 89.4796

c) Let

$$y = \frac{1}{1+x} \iff x = \frac{1-y}{y}.$$

Then we have

$$dx = \frac{1}{y^2} dy, \quad x = 0 \Rightarrow y = 1, \quad x \rightarrow \infty \Rightarrow y \rightarrow 0.$$

Which gives,

$$\int_0^\infty g(x) dx = \int_1^0 g\left(\frac{1-y}{y}\right) \frac{1}{y^2} dy = \int_0^1 g\left(\frac{1-y}{y}\right) \frac{1}{y^2} dy.$$

Define

$$\tilde{g}(u) = g\left(\frac{1-u}{u}\right) \frac{1}{u^2},$$

$\tilde{g}(u)$:

$$g(x) = \frac{x}{(1+x^2)^2}, \quad x = \frac{1-u}{u}.$$

Then

$$1+x^2 = 1 + \left(\frac{1-u}{u}\right)^2 = \frac{u^2 + (1-u)^2}{u^2}.$$

Substituting into $\tilde{g}(u)$,

$$\tilde{g}(u) = \frac{\frac{1-u}{u}}{\left(\frac{u^2 + (1-u)^2}{u^2}\right)^2} \cdot \frac{1}{u^2} = \frac{u(1-u)}{(u^2 + (1-u)^2)^2}.$$

$$\hat{I} = \frac{1}{n} \sum_{i=1}^n \tilde{g}(U_i) = \frac{1}{n} \sum_{i=1}^n \frac{U_i(1-U_i)}{(U_i^2 + (1-U_i)^2)^2}, \quad U_i \sim \text{Unif}(0, 1).$$

pseudo code:

```

1  sum = 0
2  repeat i = 1..n:
3      U = Uniform(0,1)
4      denom = U^2 + (1-U)^2
5      val = (U*(1-U)) / (denom^2)
6      sum += val
7  output (1/n) * sum

```

Converting this pseudo-code to python code and running for 1000 trials gives the estimate of the integral to be: 0.4983

d) It can be seen that

$$f_{X,Y}(x,y) = e^{-(x+y)} = e^{-x} \cdot e^{-y},$$

this gives $X, Y \stackrel{i.i.d.}{\sim} \text{Exp}(1)$. We can then define:

$$h(x,y) = \sin(xy) \mathbf{1}_{\{y \leq x^2\}}.$$

$$I = \mathbb{E}[h(X,Y)], \quad \text{where } X, Y \sim \text{Exp}(1).$$

pseudo code:

```
1  repeat i = 1..n:  
2      U = Uniform(0,1)  
3      V = Uniform(0,1)  
4      X = -log(U)  
5      Y = -log(V)  
6  if Y_i <= X_i:  
7      h(x_i, y_i) = sin(X_i, Y_i)  
8  else:  
9      h(x_i, y_i) = 0  
10 output (1/n) * [sum(h(x,y))]
```

Converting this pseudo-code to python code and running for 1000 trials gives the estimate of the integral to be: 0.1538